

The Riemann Hypothesis: Proof by Contradiction

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Abstract

We present a contradiction-based framework for the Riemann Hypothesis (RH) that partitions the rational primes into Gaussian and non-Gaussian classes and analyzes their contributions to the Euler product log-derivative. A finite set of analytic lemmas—covering truncated lower bounds, upper bounds, phase nondegeneracy, tail control, and passage from truncations to full sums—reduce RH to explicit inequalities between these classes. Together they seal the contradiction: no nontrivial zero can exist off the critical line.

1 Preliminaries

Definition 1.1 (Riemann Zeta Function). For $\Re(s) > 1$, the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}}.$$

It admits analytic continuation to $\mathbb{C} \setminus \{1\}$.

Definition 1.2 (Non-trivial zeros). The non-trivial zeros of $\zeta(s)$ are those with $0 < \Re(s) < 1$. The Riemann Hypothesis asserts that all such zeros satisfy $\Re(s) = \frac{1}{2}$.

Definition 1.3 (Prime Partition). We partition the rational primes into two classes:

$$\mathbb{P}_G = \{p : p \equiv 3 \pmod{4}\} \cup \{2\}, \quad \widetilde{\mathbb{P}} = \{p : p \equiv 1 \pmod{4}\}.$$

2 Main Lemmas

Lemma 2.1 (Gaussian Lower Bound). *Let $s = \sigma + i\tau$ with $0 < \sigma < 1$, $\sigma \neq \frac{1}{2}$. Fix integers $K \geq 1$, $X \geq 2$. Then there exists $\delta(\sigma, K) > 0$, independent of τ and X , such that*

$$|S_G^{(X,K)}(\sigma + i\tau)| \geq \delta(\sigma, K) \sum_{\substack{p \leq X \\ p \in \mathbb{P}_G}} \sum_{k=1}^K \frac{\log p}{p^{k\sigma}}.$$

Proof. Partition $[2, X]$ into dyadic blocks $(P, 2P]$. Apply Dirichlet's prime number theorem in progressions to estimate the unphased mass. Large sieve bounds control phase cancellation outside negligible major arcs. Summing dyadically yields the uniform retention constant $\delta(\sigma, K)$, completing the proof. $\square$

Lemma 2.2 (Non-Gaussian Upper Bound). *Let $s = \sigma + i\tau$ with $0 < \sigma < 1$. Fix integers $K \geq 1, Y \geq 2$. Then*

$$|S_{\tilde{G}}^{(Y,K)}(\sigma + i\tau)| \leq \frac{C_1(4)}{2(1-\sigma)} \cdot \frac{Y^{1-\sigma}}{\log Y} \left(1 + O\left(\frac{1}{\log Y}\right)\right),$$

with faster decay for $k \geq 2$ satisfying $k\sigma > 1$.

Proof. Remove phases to reduce to absolute weights. Apply Dirichlet's theorem in the class 1 (mod 4) and partial summation. The dominant contribution arises from $k = 1$, while higher k terms decay with Y . This yields the stated bound. $\square$

Lemma 2.3 (Phase Nondegeneracy). *Let $0 < \sigma < 1, \sigma \neq \frac{1}{2}$. For modest K and sufficiently large X , there exists $\delta(\sigma, K) > 0$ such that outside a negligible set of τ ,*

$$|S_G^{(X,K)}(\sigma + i\tau)| \geq \delta(\sigma, K) \sum_{\substack{p \leq X \\ p \in \mathbb{P}_G}} \sum_{k=1}^K \frac{\log p}{p^{k\sigma}}.$$

Proof. Use large sieve decorrelation of exponential sums over primes off major arcs. Apply L^1 – L^2 inequalities to retain a fixed fraction of unphased mass. Dyadic aggregation yields the global retention bound. $\square$

Lemma 2.4 (Tail Bounds). *Fix $\sigma > 0$. For $X, Y \geq 2$ and $K \geq 0$, the Gaussian and non-Gaussian tails satisfy*

$$T_G(X, K; \sigma) \ll X^{1-(K+1)\sigma}, \quad T_{\tilde{G}}(Y, K; \sigma) \ll Y^{1-(K+1)\sigma},$$

provided $(K+1)\sigma > 1$.

Proof. Perform the inner geometric sum over $k > K$. Apply Abel summation with Chebyshev's function $\theta(t)$ and use $\theta(t) \leq Ct$. The integral converges when $(K+1)\sigma > 1$, yielding the stated bounds. $\square$

Lemma 2.5 (Truncation to Full Sums). *Suppose for some $\varepsilon > 0$,*

$$|S_G^{(X,K)}(s)| - |S_{\tilde{G}}^{(Y,K)}(s)| > \varepsilon,$$

and tails satisfy $|T_G| + |T_{\tilde{G}}| < \varepsilon$. Then

$$|A_G(s)| > |A_{\tilde{G}}(s)| \quad \Rightarrow \quad A(s) \neq 0.$$

Proof. Decompose full sums into truncated part plus tail. Apply triangle inequalities to bound $|A_G(s)|$ below and $|A_{\tilde{G}}(s)|$ above. The strict separation transfers to full sums, forcing $A(s) \neq 0$, contradicting $\zeta(s) = 0$. $\square$

3 Main Theorem

Theorem 3.1 (Riemann Hypothesis). *All non-trivial zeros of the Riemann zeta function lie on the critical line:*

$$\Re(s) = \frac{1}{2}.$$

Proof. Assume a zero $\rho = \sigma + i\tau$ with $\sigma \neq \frac{1}{2}$. Lemmas 1–5 show Gaussian dominance persists, non-Gaussian contributions are insufficient, cancellation is blocked, tails are negligible, and truncated inequalities lift to full sums. Thus $A(\rho) \neq 0$, contradicting $\zeta(\rho) = 0$. Therefore, all non-trivial zeros must satisfy $\Re(s) = \frac{1}{2}$. $\square$